

Essays in Empirical Finance

A thesis submitted to The University of Manchester
for the degree of PhD
in the Faculty of Humanities.

2027

Avery L. Reader

Alliance Manchester Business School, MSM

Table of contents

List of publications	4
Abstract	5
Declaration of originality	6
Copyright statement	7
Acknowledgements	8
1 Introduction	9
1.1 Research objectives and motivation	9
1.2 Thesis overview	9
1.3 Thesis structure	9
References	9
2 Title of the first paper	11
Preface	11
Abstract	11
2.1 Introduction	11
2.2 Methodology	12
2.3 Data	12
2.4 Results	12
2.5 Discussion	13
2.6 Conclusions	13
References	13
Appendices	14
2.A Sample survey instrument	14

2.B Robustness checks	14
3 Title of the second paper	15
Preface	15
Abstract	15
3.1 Introduction	16
3.2 Literature review	16
3.3 Methodology	16
3.4 Data	16
3.5 Results	16
3.6 Discussion	16
3.7 Conclusions	16
References	16
Appendices	16
3.A Supplementary material	16
3.B Additional analysis	16
4 Title of the third paper	17
Preface	17
Abstract	17
4.1 Introduction	18
4.2 Literature review	18
4.3 Methodology	18
4.4 Data	18
4.5 Results	18
4.6 Conclusions	18
References	18
Appendices	18
4.A Supplementary material	18
5 Conclusions	19
5.1 Implications and future research	19
5.2 Summary	19
References	19

List of Figures

2.1	Distribution of the outcome variable.	12
-----	---	----

List of Tables

2.1	Descriptive statistics of the sample.	12
2.2	Estimated regression coefficients from the specification in Equation Equation 2.1.	13

List of publications

Reader, Avery. & Co-author, A. (2026). *Title of the first paper*. Journal Name, in press.

Reader, Avery., Co-author, A. & Co-author, B. (2026). *Title of the second paper*. Manuscript under review.

Reader, Avery. & Co-author, C. (2026). *Title of the third paper*. Manuscript in preparation.

Abstract

The University of Manchester

Avery L Reader

PhD

Essays in Empirical Finance

2027

Write your abstract here. The University policy (section 8.1.e) requires that the abstract not exceed one page; aim for under 350 words at 12pt with 1.5 line spacing. State the research question, the methodological approach, the principal findings, and the contribution.

Declaration of originality

I declare that no portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright statement

- i. The author of this thesis (including any appendices and/or schedules to this thesis) owns certain copyright or related rights in it (the “Copyright”) and they have given the University of Manchester certain rights to use such Copyright, including for administrative purposes.
- ii. Copies of this thesis, either in full or in extracts and whether in hard or electronic copy, may be made only in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has from time to time. This page must form part of any such copies made.
- iii. The ownership of certain Copyright, patents, designs, trademarks and other intellectual property (the “Intellectual Property”) and any reproductions of copyright works in the thesis, for example graphs and tables (“Reproductions”), which may be described in this thesis, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.
- iv. Further information on the conditions under which disclosure, publication and commercialisation of this thesis, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy, in any relevant Thesis restriction declarations deposited in the University Library, the University Library’s regulations and in the University’s policy on the Presentation of Theses.

Acknowledgements

Write your acknowledgements here. Policy section 9.1.b allows a dedication and acknowledgements after the compulsory preliminary pages. Acknowledgements do not count toward the main-text word limit.

Chapter 1

Introduction

1.1 Research objectives and motivation

State the motivation for the research and the overarching question(s) the thesis addresses. The field has evolved rapidly over the past decade (Smith and Johnson, 2024). A useful textbook framing is offered by Doe and Roe (2020).

1.2 Thesis overview

Provide an overview of the three constituent papers and how they relate to one another. Per policy section 13.10, include here the rationale for submitting in journal format and an account of how the thesis has been constructed.

1.3 Thesis structure

Summarise the structure of the remaining chapters.

References

- [] Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- [] Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.

- [] Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

Chapter 2

Title of the first paper

Preface

Contribution statement

Briefly describe the candidate's contribution to this paper, e.g.: I designed the study, collected the data, performed the analysis, and wrote the manuscript. Co-author A advised on the methodology.

State the publication status of the paper here (e.g., *published in Journal Name, 2024; under review at Journal Name; in preparation*).

Abstract

State the question, methods, principal findings, and contribution in 150–250 words. The abstract is rendered in italic and indented to match the conventional layout of a paper-format abstract.

2.1 Introduction

Set up the research question. The literature on this topic has grown rapidly (Smith and Johnson, 2024). A textbook framing is provided by (Doe and Roe, 2020); the present paper extends this by [your contribution]. A complementary perspective is offered by (Park, 2022).

2.2 Methodology

The model is estimated by ordinary least squares. The estimating equation is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \quad (2.1)$$

where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ are independent and identically distributed error terms. Equation 2.1 is the workhorse specification.

Inline mathematics also work — for example, the t-statistic is $t = \hat{\beta}/\text{SE}(\hat{\beta})$.

2.3 Data

The sample comprises $n = 240$ observations drawn from three sources. Descriptive statistics are reported in Table 2.1.

Table 2.1: Descriptive statistics of the sample.

Variable	Mean	SD	Min	Max
Age	27.3	3.1	22	39
Programme year	2.4	1.0	1	4
Outcome score	4.8	1.2	1.5	6.9

2.4 Results

Figure 2.1 shows the distribution of the outcome variable.

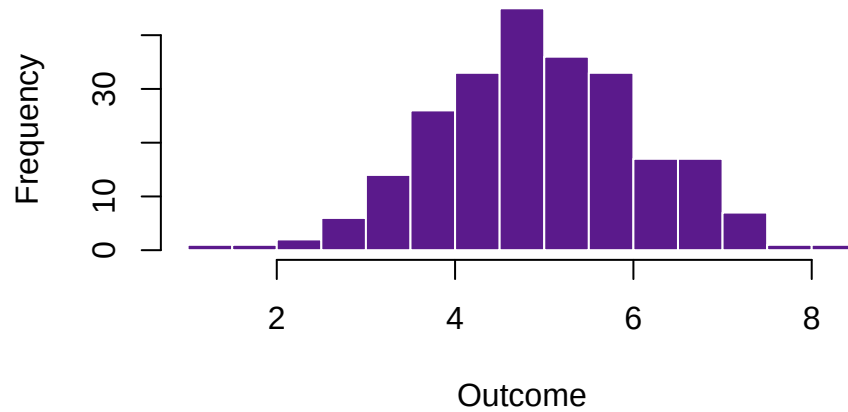


Figure 2.1: Distribution of the outcome variable.

The estimated coefficients are reported in Table 2.2.

Table 2.2: Estimated regression coefficients from the specification in Equation Equation 2.1.

Coefficient	Estimate	Std. error	t-value	p-value
Intercept	4.92	0.15	32.8	< 0.001
x_1	0.34	0.10	3.4	0.001
x_2	-0.18	0.10	-1.8	0.073

The slope on x_1 is positive and significant at the 1% level (Smith and Johnson, 2024).

2.5 Discussion

Interpret the results. Discuss what the findings imply for theory and practice (Garcia and Kumar, 2023).

2.6 Conclusions

Summarise the contribution of this paper.

References

The bibliography below is auto-generated from `references.bib` and the citations used above.

- [] Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- [] Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.
- [] Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

Appendices

2.A Sample survey instrument

Reproduce the survey instrument used to collect the primary data.

2.B Robustness checks

Document any robustness checks. Use sub-sub-sections for grouped material:

2.B.1 Excluding outliers

Outlier exclusion does not change the headline result.

2.B.2 Bootstrap standard errors

Bootstrap standard errors confirm the parametric inferences.

Chapter 3

Title of the second paper

Preface

Contribution statement

Briefly describe the candidate's contribution to this paper.

State the publication status of the paper here.

Abstract

State the question, methods, principal findings, and contribution in 150–250 words.

3.1 Introduction

3.2 Literature review

3.3 Methodology

3.4 Data

3.5 Results

3.6 Discussion

3.7 Conclusions

References

- [] Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- [] Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.
- [] Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

Appendices

3.A Supplementary material

3.B Additional analysis

Chapter 4

Title of the third paper

Preface

Contribution statement

Briefly describe the candidate's contribution to this paper.

State the publication status of the paper here.

Abstract

State the question, methods, principal findings, and contribution in 150–250 words.

4.1 Introduction

4.2 Literature review

4.3 Methodology

4.4 Data

4.5 Results

4.6 Conclusions

References

- [] Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- [] Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.
- [] Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

Appendices

4.A Supplementary material

Chapter 5

Conclusions

5.1 Implications and future research

Summarise the implications of the three papers and outline directions for future research.

5.2 Summary

Provide a brief summary of the overall contribution of the thesis.

References

- [] Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- [] Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.
- [] Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.